PRODUCT USER MANUAL

For Arctic Ocean Physical and BGC Analysis and Forecasting Products

SEALEVEL_GLO_PHY_MDT_008_063

SEALEVEL_MED_PHY_MDT_L4_STATIC_008_066

SEALEVEL_BLK_PHY_MDT_L4_STATIC_008_067

SEALEVEL_EUR_PHY_MDT_L4_STATIC_008_070

Issue: 4.0

Contributors : Marie-Isabelle Pujol (CLS)

Approval Date : November 2024

MERCATOR OCEAN
INTERNATIONAL
2 avenue de l'aérodrome de Montaudran, 31400, TOULOUSE, FRANCE
Tél : +33 5 61 39 38 02 - Fax : +33 5 61 39 38 99
Société civile de droit français au capital de 2 000 000 € - 522 911 577 RCS Toulouse - SIRET 522 911 577 00016

marine.copernicus.eu
mercator-ocean.eu

RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2021/01/15	all	Initial version. Merge PUM for Global MDT with additional information for regional MDTs	M-I Pujol	M-I Pujol
2.0	2021/10/09		Update of SEALEVEL_GLO_PHY_MDT_008_063 product	M-I Pujol	M-I Pujol
3.0	2023/06/15	all	Use new template Update of SEALEVEL_GLO_PHY_MDT_008_063 product	M-I Pujol	M-I Pujol
4.0	2024/05/31		Add of SEALEVEL_EUR_PHY_MDT_L4_STATIC_008_070 product	M-I Pujol	M-I Pujol

Table of Contents

GLOSSARY AND ABBREVIATIONS *4*

DATA ACCESS **5**

I INTRODUCTION *6*

I.1 Summary **6**

I.2 History of changes **7**

I.3 Copernicus Marine Service and Aviso+ Disseminations **7**

II DESCRIPTION OF THE PRODUCT SPECIFICATION *8*

II.1 General Information **8**

II.2 Details of datasets **11**

II.3 Additional Information on parameters **12**

III FILE FORMAT *13*

IV FILES NOMENCLATURE *14*

IV.1 Nomenclature of files when downloaded through the Subset Service **14**

IV.2 Nomenclature of original files **14**

IV.3 File Format **15**

IV.4 File size **15**

IV.5 Structure of files **16**

V REFERENCES *17*

VI ANNEX : **17**

Differents Heigths used in altimetry ***20***

GLOSSARY AND ABBREVIATIONS

CF	Climate Forecast (convention for NetCDF)
CMEMS	Copernicus Marine Environment Monitoring Service
DUACS	Data Unification and Altimeter Combination System
ECMWF	European Centre for Medium Range Weather forecast
L3	Level-3 (along-track altimeter products)
MDT	Mean Dynamic Topography
NetCDF	Network Common Data Form
PUM	Product User Manual
QUID	Quality Information Document
SLA	Sea Level Anomaly
SQO	Synthesis Quality Overview
TAC	Thematic Assembly Centre

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

I.1 Summary

This document is the user manual for the Mean Dynamic Topography (MDT), used as auxiliary product in the altimeter measurement DUACS processing to retrieve an Absolute Dynamic Topography above the geoid.

The MDT_N is the temporal mean of the SSH above the Geoid over a period N. (see Annex for details). The MDT described by this document are referenced to the reference period (N) of 20-year covering from 1993 to 2012. This is consistent with the reference time period also used in the SSALTO DUACS products.

The processing steps are described in the Quality Information Document (QUID) ; Product quality information are given in the Synthesis Quality Overview (SQO) Document: see in **Erreur ! Source du renvoi introuvable.**)

I.2 History of changes

Date	Description of changes and impacted product
2021/11	The version of the global MDT (product 008_063) was updated with the hybrid MSS CNES-CLS-2018/CMEMS-2020
2023/11	The version of the global MDT (product 008_063) was updated with the hybrid MSS CNES-CLS-2022/CMEMS-2020
2024/05	The MDT for the Europe area was added

I.3 Copernicus Marine Service and Aviso+ Disseminations

The **MDT** for global ocean is available via the Aviso+ dissemination (with registration) and is available via the https://www.aviso.altimetry.fr/en/data/products/auxiliary-products/mdt.html . It is also disseminated by Copernicus Marine Service.

II DESCRIPTION OF THE PRODUCT SPECIFICATION

II.1 General Information

Product Lines	SEALEVEL_GLO_PHY_MDT_008_063
Geographical coverage	Global
Variables	• latitude • longitude • mdt • u • v • err_mdt • err_u • err_v
Product Type	Multi Year - Static
Available time series	N/A
Temporal resolution	N/A
Target delivery time	Once
Delivery mechanism	Marine Data Store
Horizontal resolution	1/8° x 1/8° grid
Number of vertical levels	surface only
Format	NetCDF CF1.6

Table 1: Global MDT product

Product Lines	SEALEVEL_MED_PHY_MDT_L4_STATIC_008_066
Geographical coverage	Mediterranean Sea
Variables	• latitude • longitude • mdt • u • v • err_mdt • err_u • err_v
Product Type	Multi Year - Static
Available time series	N/A
Temporal resolution	N/A
Target delivery time	Once
Delivery mechanism	Marine Data Store
Horizontal resolution	1/24° x 1/24° grid
Number of vertical levels	surface only
Format	NetCDF CF1.6

Table 2 : Regional Mediterranean MDT product

Product Lines	SEALEVEL_BLK_PHY_MDT_L4_STATIC_008_067
Geographical coverage	Black Sea
Variables	• latitude • longitude • mdt • u • v • err_mdt • err_u • err_v
Product Type	Multi Year - Static
Available time series	N/A
Temporal resolution	N/A
Target delivery time	Once
Delivery mechanism	Marine Data Store
Horizontal resolution	1/16° x 1/16° grid
Number of vertical levels	surface only
Format	NetCDF CF1.6

Table 3 : Regional Black Sea MDT product

Product Lines	SEALEVEL_EUR_PHY_MDT_L4_STATIC_008_070
Geographical coverage	Europe
Variables	• latitude • longitude • mdt • u • v • err_mdt • err_u • err_v
Product Type	Multi Year - Static
Available time series	N/A
Temporal resolution	N/A
Target delivery time	Once
Delivery mechanism	Marine Data Store
Horizontal resolution	1/16° x 1/16° grid
Number of vertical levels	surface only

Format	NetCDF CF1.6

Table 4 : Regional Europe MDT product

II.2 Details of datasets

SEALEVEL_GLO_PHY_MDT_008_063 **SEALEVEL_MED_PHY_MDT_L4_STATIC_008_066** **SEALEVEL_BLK_PHY_MDT_L4_STATIC_008_067** **SEALEVEL_EUR_PHY_MDT_L4_STATIC_008_070**
Datasets: cnes_obs-sl_glo_phy-mdt_my_0.125deg_P20Y: global MDT cmems_obs-sl_med_phy-mdt_my_l4-0.0417deg_P20Y: Mediterranean Sea MDT cmems_obs-sl_blk_phy-mdt_my_l4-0.0625deg_P20Y: Black Sea MDT cmems_obs-sl_eur_phy-mdt_my_l4-0.0625deg_P20Y: Europe MDT **Variables name in the NetCDF file , unit long_name & Standard_name**
mdt **[m]** **mean dynamic topography** **sea_surface_height_above_geoid**
u **[m/s]** **Mean eastward geostrophic velocity** **surface_geostrophic_eastward_sea_water_velocity**
v **[m/s]** **Mean northward geostrophic velocity** **surface_geostrophic_northward_sea_water_velocity**
err_mdt **[m]** **mean dynamic topography error** **sea_surface_height_above_geoid standard_error**
err_u **[m]** **mean eastward geostrophic velocity error** **surface_geostrophic_eastward_sea_water_velocity standard_error**
err_v **[m]** **mean northward geostrophic velocity error** **surface_geostrophic_northward_sea_water_velocity standard_error**

Table 2: list of the datasets and variable names and unit for the SEALEVEL_GLO_PHY_MDT_008_063, SEALEVEL_MED_PHY_MDT_L4_STATIC_008_066, SEALEVEL_BLK_PHY_MDT_L4_STATIC_008_067 products

II.3 Additional Information on parameters

mdt	Mean Dynamic Topography referenced to the [1993, 2012] period (see Annex) The mdt is computed using multivariate optimal interpolation method (see associated QUID document for details).
u & v	Mean eastward & northward geostrophic velocity. The u & v components are computed using multivariate optimal interpolation method (see associated QUID document for details).
err_mdt	Formal mapping error associated to the mdt field. This formal error corresponds to a local minimum in the least squares senses
err_u & err_v	Formal mapping error associated to the u & v fields This formal error corresponds to a local minimum in the least squares senses

III FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link: What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:

how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV FILES NOMENCLATURE

IV.1 Nomenclature of files when downloaded through the Subset Service

The files nomenclature when downloaded through the Copernicus Marine Web Portal Subset is based on product dataset name and a numerical reference related to the request date on the portal.

The scheme is: **datasetname_nnnnnnnnnnnnn.nc**

where:

- **datasetname**: as described previously
- **nnnnnnnnnnnnn**: 13 digits integer corresponding to the current time (download time) in milliseconds since January 1, 1970 midnight UTC.
- **.nc**: standard NetCDF filename extension.

IV.2 Nomenclature of original files

The files nomenclature is the following:

mdt_<mdt_name>_<area>.<format>

where the fileds in "<>" are described below:

mdt_name	hybrid_cnes_cls22_cmems2020 cmems_2020 cmems_2024	Hybrid MDT combining CNES_CLS 2022 solution over global area and CMEMS_2020 solutions for Mediterranean and Black Sea area CMEMS 2020 MDT CMEMS 2024 MDT
area	global med blk europe	global geographic coverage product Mediterranean Sea coverage Black Sea coverage Europe area coverage
format	.nc	NetCdf CF1.6

Example: mdt_hybrid_cnes_cls22_cmems2020_global.nc ; mdt_cmems_2020_med.nc

IV.3 File Format

The products are stored using the NetCDF format.

The NetCDF SEA LEVEL TAC files are based on the attribute data tags defined by the Cooperative Ocean/Atmosphere Research Data Service (COARDS) and Climate and Forecast (CF) metadata conventions. The CF convention generalises and extends the COARDS convention but relaxes the COARDS constraints on dimension and order and specifies methods for reducing the size of datasets.

In addition to these conventions, the files are using a common structure and semantic as described below:

[To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF]

IV.4 File size

PRODUCT/DATASET NAME	FILE NAME	DIMENSION [MB]
		NetCDF-4 Compressed
SEALEVEL_GLO_PHY_MDT_008_063/cnes_obs-sl_glo_phy-mdt_my_0.125deg_P20Y	hybrid_cnes_cls22_cmems2020.nc	100
SEALEVEL_MED_PHY_MDT_L4_STATIC_008_066/cmems_obs-sl_med_phy-mdt_my_l4-0.0417deg_P20Y	mdt_cmems_2020_med.nc	0.145
SEALEVEL_BLK_PHY_MDT_L4_STATIC_008_067/cmems_obs-sl_blk_phy-mdt_my_l4-0.0625deg_P20Y	mdt_cmems_2020_blk.nc	< 0.1
SEALEVEL_EUR_PHY_MDT_L4_STATIC_008_070/cmems_obs-sl_eur_phy-mdt_my_l4-0.0625deg_P20Y	mdt_cmems_2024_europe.nc	22

IV.5 Structure of files

Examples of the header of output NetCDF files are inserted in annex, for each dataset.

V REFERENCES

Quality Information Document (QUID) :
https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-SL-QUID-008-063.pdf

Synthesis Quality Overview (SQO) Document:
https://marine.copernicus.eu/documents/QUID/CMEMS-SL-SQO-008-063.pdf.

VI ANNEX :

An example output NetCDF file header for SEALEVEL_MED_PHY_MDT_L4_STATIC_008_066 is given below:

```
netcdf mdt_cmems_2020_med {
dimensions:
    nv = 2 ;
    latitude = 434 ;
    longitude = 1014 ;
    time = 1 ;
variables:
    float time(time) ;
        time:axis = "T" ;
        time:bounds = "climatology_bnds" ;
        time:calendar = "gregorian" ;
        time:long_name = "Time" ;
        time:standard_name = "time" ;
        time:units = "days since 1950-01-01 00:00:00 UTC" ;
    float climatology_bnds(time, nv) ;
    int crs ;
        crs:comment = "This is a container variable that describes the grid_mapping used by the data in this file. This variable does not
contain any data; only information about the geographic coordinate system." ;
        crs:grid_mapping_name = "latitude_longitude" ;
        crs:inverse_flattening = 298.257 ;
        crs:semi_major_axis = 6378136.3 ;
    int nv(nv) ;
        nv:comment = "Vertex" ;
        nv:units = "1" ;
    float lat_bnds(latitude, nv) ;
        lat_bnds:comment = "latitude values at the north and south bounds of each pixel." ;
        lat_bnds:units = "degrees_north" ;
    float lon_bnds(longitude, nv) ;
        lon_bnds:comment = "longitude values at the west and east bounds of each pixel." ;
        lon_bnds:units = "degrees_east" ;
    float latitude(latitude) ;
        latitude:bounds = "lat_bnds" ;
        latitude:long_name = "Latitude" ;
        latitude:standard_name = "latitude" ;
        latitude:units = "degrees_north" ;
        latitude:axis = "Y" ;
        latitude:valid_min = 29.02084f ;
        latitude:valid_max = 47.0625f ;
    float longitude(longitude) ;
        longitude:bounds = "lon_bnds" ;
```

```
        longitude:long_name = "Longitude" ;
        longitude:standard_name = "longitude" ;
        longitude:units = "degrees_east" ;
        longitude:axis = "X" ;
        longitude:valid_min = -6.062497f ;
        longitude:valid_max = 36.14583f ;
    int mdt(time, latitude, longitude) ;
        mdt:_FillValue = -2147483647 ;
        mdt:long_name = "Mean dynamic topography" ;
        mdt:standard_name = "sea_surface_height_above_geoid" ;
        mdt:scale_factor = 0.0001 ;
        mdt:grid_mapping = "crs" ;
        mdt:units = "m" ;
        mdt:ancillary_variables = "mdt_error" ;
        mdt:coordinates = "longitude latitude" ;
    int u(time, latitude, longitude) ;
        u:_FillValue = -2147483647 ;
        u:long_name = "Mean eastward geostrophic velocity" ;
        u:standard_name = "surface_geostrophic_eastward_sea_water_velocity" ;
        u:scale_factor = 0.0001 ;
        u:grid_mapping = "crs" ;
        u:units = "m/s" ;
        u:ancillary_variables = "u_error" ;
        u:coordinates = "longitude latitude" ;
    int v(time, latitude, longitude) ;
        v:_FillValue = -2147483647 ;
        v:long_name = "Mean northward geostrophic velocity" ;
        v:standard_name = "surface_geostrophic_northward_sea_water_velocity" ;
        v:scale_factor = 0.0001 ;
        v:grid_mapping = "crs" ;
        v:units = "m/s" ;
        v:ancillary_variables = "v_error" ;
        v:coordinates = "longitude latitude" ;
    int err_mdt(time, latitude, longitude) ;
        err_mdt:_FillValue = -2147483647 ;
        err_mdt:long_name = "Mean dynamic topography error" ;
        err_mdt:standard_name = "sea_surface_height_above_geoid standard_error" ;
        err_mdt:scale_factor = 0.0001 ;
        err_mdt:grid_mapping = "crs" ;
        err_mdt:units = "m" ;
        err_mdt:coordinates = "longitude latitude" ;
    int err_u(time, latitude, longitude) ;
        err_u:_FillValue = -2147483647 ;
        err_u:long_name = "Mean eastward geostrophic velocity error" ;
        err_u:standard_name = "surface_geostrophic_eastward_sea_water_velocity standard_error" ;
        err_u:scale_factor = 0.0001 ;
        err_u:grid_mapping = "crs" ;
        err_u:units = "m/s" ;
        err_u:coordinates = "longitude latitude" ;
    int err_v(time, latitude, longitude) ;
        err_v:_FillValue = -2147483647 ;
        err_v:long_name = "Mean northward geostrophic velocity error" ;
        err_v:standard_name = "surface_geostrophic_northward_sea_water_velocity standard_error" ;
        err_v:scale_factor = 0.0001 ;
        err_v:grid_mapping = "crs" ;
        err_v:units = "m/s" ;
        err_v:coordinates = "longitude latitude" ;

// global attributes:
        :Conventions = "CF-1.6" ;
        :Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;
        :history = "2020-03-08T00:00:00Z : Creation " ;
        :contact = "servicedesk.cmems@mercator-ocean.eu, smulet@groupcls.com, sjousset@groupcls.com" ;
        :creator_email = "servicedesk.cmems@mercator-ocean.eu " ;
        :creator_url = "http://marine.copernicus.eu " ;
        :creator_name = "CMEMS" ;
```

:date_created = "2020-03-08T00:00:00Z" ;
:date_issued = "2020-03-08T00:00:00Z " ;
:date_modified = "2020-03-08T00:00:00Z " ;
:geospatial_lat_min = 29.02084 ;
:geospatial_lat_max = 47.0625 ;
:geospatial_lon_min = -6.062497 ;
:geospatial_lon_max = 36.14583 ;
:geospatial_vertical_min = "0.0" ;
:geospatial_vertical_max = "0.0" ;
:geospatial_lat_units = "degrees_north" ;
:geospatial_lon_units = "degrees_east" ;
:geospatial_lat_resolution = 0.04166 ;
:geospatial_lon_resolution = 0.04166 ;
:geospatial_vertical_positive = "down" ;
:geospatial_vertical_resolution = "point" ;
:geospatial_vertical_units = "m" ;
:institution = "CLS, CMEMS" ;
:keywords = "Oceans > Ocean mean dynamic topography " ;
:keywords_vocabulary = "NetCDF COARDS Climate and Forecast Standard Names" ;
:processing_level = "L4" ;
:product_version = "MDT CMEMS_2020_MED v1.0" ;
:references = "Jousset et al, OSTST 2020" ;
:source = "Altimetry measurements, gravimetry (icluding GOCE and GRACE) and in-situ data." ;
:summary = "Mean Dynamic Topography of the Mediterranean Sea calculated from the combination of altimetry, gravimetry (including GOCE and GRACE) and in-situ data. The reference time-period is 1993-2012." ;
:title = "MDT CMEMS_2020_MED " ;
:standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata Convention Standard Name Table v37" ;
:time_coverage_duration = "P20Y" ;
:time_coverage_start = "1993-01-01T00:00:00Z" ;
:time_coverage_resolution = "P20Y" ;
:time_coverage_end = "2013-01-01T00:00:00Z" ;
}

DIFFERENTS HEIGTHS USED IN ALTIMETRY

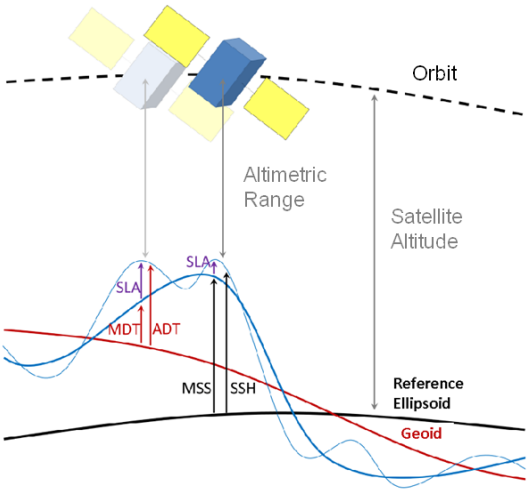

Figure 1: Altimetry principle

- The Altimetry gives access to the Sea Surface Height (SSH) above the reference ellipsoid (see figure 1)

SSH = Orbit - Altimetric Range

- The **Mean Sea Surface (MSS)** is the temporal mean of the **SSH** over a period N.

  It is a mean surface above the **ellipsoïd** and it includes the **Geoid**. See the detailed computation in QUID document

$$\mathbf{MSS_N = \langle SSH \rangle_N}$$

- The dynamical part of the signal: **Sea Level Anomaly (SLA)** is deduced from the **SSH** using a **Mean Sea Surface (MSS):**

$$\mathbf{SLA_N = SSH - MSS_N}$$

- The **Mean Dynamic Topography (MDT)** is the temporal mean of the **SSH** above the **Geoid** over a period N.

$$\mathbf{MDT_N = MSS_N - Geoid}$$

The dynamical part of the absolute signal: **Absolute Dynamic Topography (ADT)** is deduced from the **SLA** using a **Mean Dynamic Topography (MDT):**

$$\mathbf{ADT = SLA_N + MDT_N = SSH - MSS_N + MDT_N}$$